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End-to-end production simulation

Document: SIM-EEG-018 **Revision:** A **Date:** 2026-09-02 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Licence:** CC BY-SA 4.0
Generated by: `tools/simulate_production.py`, from `tools/design.py` and the files actually present in the package

What this is, and what it is not

This report is a dry run of the manufacturing route against the data package. At each station it asks what the operator needs, checks that the package contains it, and where the station has an acceptance limit it computes that limit from component values instead of quoting it. It is re-run by `tools/simulate_production.py` and must be clean before the package is released.

It cannot find what only a real build finds: solderability, fit, ergonomics, electromagnetic behaviour, or whether a participant can actually put the helmet on. Nothing in this package has been manufactured or measured.

Result: 193 checks passed, 0 failed, 7 known open items.

A *failure* means the package does not support the step. An *open item* means it does, and somebody still has to take a decision that is recorded where it belongs. The open items are:

Station	Item	Detail
01 bare-board fabrication	static IRAM reports full, with one byte free	1 byte(s) free of 16,384 on the linked image. If that is the real limit then the next function marked IRAM_ATTR fails the link with an error naming a section and not a cause, and this design has more interrupt work coming (E-13's tone scheduler, E-12's onset detector). Turning off the SPI-slave and gptimer ISRs – neither of which this firmware uses – did not move the figure by a single byte, so it is not those. Either the pool is genuinely full, or esp_idf_size is reporting against a fixed 16 KB window that is not the limit on an ESP32-S3 with octal SPIRAM and XIP. Reading the linker map against hardware settles it; guessing at sdkconfig from here does not

Station	Item	Detail
01 bare-board fabrication	the v1 HM-01 mesh is two disconnected bodies	measured with trimesh: it is watertight, which MANIFEST.json records, and it is not connected, which nothing recorded. It is a carried-over form study rather than a released printable part – the parametric channel runs HM-01P-A/B/C were measured from it – but a frame that is two shells is one more reason the redraw of PARTS-EEG-019 OA-1 has to happen
01 bare-board fabrication	there is no fabrication data for the current board	Rev B is withdrawn from fabrication (ECO-EEG-030) and Rev C is UNROUTED: its placement and routing are bought. The Gerbers, drill, DRC report and drawings this station checks are Rev B's and are kept as history. A bare board cannot be ordered. What exists for Rev C is the layout input set in kicad/RevC_layout_inputs/ and the rule sheet LAY-EEG-034
01 bare-board fabrication	the placement of Rev C has not been reviewed	A human layout engineer HAS now read the Rev B routing: he read it between 3 and 5 September 2026, declined the paid review and advised a redesign, and Rev B is withdrawn (ECO-EEG-030). What is open is the next review, not that one. RFQ-EEG-002A is re-scoped to the review of the layout contractor's PLACEMENT of Rev C, taken at the placement confirmation gate and before routing is confirmed, as a written findings list by net, pad pair and coordinate. There is no placement to review yet
09 functional test TST-EEG-004	E-27 has never been seen to light	the driver is written and the current budget is met, but no unit exists, so no light has been driven and TST-EEG-004 T11 – which reads the R/G ratio with a colorimeter – has not been run. The alternation also quantises to the FreeRTOS tick, about 250 Hz rather than exactly 240, which meets the 'above 100 Hz' E-27 is written against and is what T11 will actually measure

Station	Item	Detail
09 functional test TST-EEG-004	the two board-current figures cannot both be right	TST-EEG-004 T3 limits J13 to 150 mA while ICD-EEG-006 section 2.7 tallies about 440 mA. E-22 is met either way (20.0 h against 6.8 h), but the charger, the thermal budget and the T3 limit itself all rest on the number, and it is open item 14 of RFQ-EEG-001 Rev E. It is measured, not calculated, at T3
09 functional test TST-EEG-004	SR-01 is closed in the design and not yet signed off	S-02 is now met at 36.8 uA against 50 uA, because ECO-EEG-024 raised R1-R16 to 68 kOhm. Applying the fix the analysis pointed to is not the same as having it approved: the electrical safety reviewer of RISK-EEG-011 section 7 owns SR-01 and that review has not started.

Station by station

Station 00 purchase order and incoming goods

	Check	Detail
pass	carrier BOM exists	/Users/nstephane/Dev/EEG-kit/kicad/EEG-CAR-01_RevC_BOM.csv
value	BOM lines	79
pass	every purchased line carries a manufacturer part number	
pass	BOM quantity equals the placed part count	BOM 207, placed 207
value	do-not-populate	R89
pass	approved vendor list present	
pass	incoming inspection defined (quality plan)	

Station 01 bare-board fabrication

	Check	Detail
pass	EEG-CAR-01-F_Cu.gbr present	
pass	EEG-CAR-01-In1_Cu.gbr present	

	Check	Detail
pass	EEG-CAR-01-In2_Cu.gbr present	
pass	EEG-CAR-01-B_Cu.gbr present	
pass	EEG-CAR-01-F_Mask.gbr present	
pass	EEG-CAR-01-B_Mask.gbr present	
pass	EEG-CAR-01-F_Silkscreen.gbr present	
pass	EEG-CAR-01-B_Silkscreen.gbr present	
pass	EEG-CAR-01-Edge_Cuts.gbr present	
pass	EEG-CAR-01-PTH.drl present	
pass	EEG-CAR-01-NPTH.drl present	
pass	EEG-CAR-01-F_Cu.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-F_Cu.gbr uses only apertures it defines	
pass	EEG-CAR-01-F_Cu.gbr coordinates lie inside the board outline	x 2.90..148.00 y 1.90..128.50
pass	EEG-CAR-01-In1_Cu.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-In1_Cu.gbr uses only apertures it defines	
pass	EEG-CAR-01-In1_Cu.gbr coordinates lie inside the board outline	x 0.40..149.60 y 0.40..129.60
pass	EEG-CAR-01-In2_Cu.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-In2_Cu.gbr uses only apertures it defines	
pass	EEG-CAR-01-In2_Cu.gbr coordinates lie inside the board outline	x 0.40..149.60 y 0.40..129.60

	Check	Detail
pass	EEG-CAR-01-B_Cu.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-B_Cu.gbr uses only apertures it defines	
pass	EEG-CAR-01-B_Cu.gbr coordinates lie inside the board outline	x 2.70..147.09 y 3.50..127.80
pass	EEG-CAR-01-F_Mask.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-F_Mask.gbr uses only apertures it defines	
pass	EEG-CAR-01-F_Mask.gbr coordinates lie inside the board outline	x 5.00..145.94 y 3.50..126.70
pass	EEG-CAR-01-B_Mask.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-B_Mask.gbr uses only apertures it defines	
pass	EEG-CAR-01-B_Mask.gbr coordinates lie inside the board outline	x 5.00..144.00 y 17.14..125.00
pass	EEG-CAR-01-F_Silkscreen.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-F_Silkscreen.gbr uses only apertures it defines	
pass	EEG-CAR-01-F_Silkscreen.gbr coordinates lie inside the board outline	x 3.00..146.03 y 4.85..128.38
pass	EEG-CAR-01-B_Silkscreen.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-B_Silkscreen.gbr uses only apertures it defines	
pass	EEG-CAR-01-B_Silkscreen.gbr coordinates lie inside the board outline	x 17.92..61.91 y 7.00..10.60

	Check	Detail
pass	EEG-CAR-01-Edge_Cuts.gbr is a complete Gerber (FS, MO, M02)	
pass	EEG-CAR-01-Edge_Cuts.gbr uses only apertures it defines	
pass	EEG-CAR-01-Edge_Cuts.gbr coordinates lie inside the board outline	x 0.00..150.00 y 0.00..130.00
pass	EEG-CAR-01-PTH.drl is a complete Excellon file (M48, M30)	
pass	EEG-CAR-01-PTH.drl declares metric units	
pass	EEG-CAR-01-PTH.drl defines every tool it uses	undeclared: []
value	EEG-CAR-01-PTH.drl tools	0.300 mm, 0.900 mm, 1.000 mm, 1.200 mm, 1.700 mm
pass	EEG-CAR-01-NPTH.drl is a complete Excellon file (M48, M30)	
pass	EEG-CAR-01-NPTH.drl declares metric units	
pass	EEG-CAR-01-NPTH.drl defines every tool it uses	undeclared: []
value	EEG-CAR-01-NPTH.drl tools	1.500 mm, 3.200 mm
pass	non-plated hole count matches the design	drill 10, design 10
pass	no hole appears in both drill files	
value	plated holes	788
pass	the pad the DRC measures is the pad the design declares	all pads agree
pass	every pin-1 legend mark is nearest pad 1	all footprints agree
pass	IPC-D-356A netlist present	
pass	every IPC-D-356A record is the same width	widths [74]

	Check	Detail
pass	every IPC-D-356A drill diameter is one the design has	5 distinct, all matched
pass	solder paste layer supplied	
pass	every stencil aperture is one ASM-EEG-007 section 2.3 specifies	7 distinct apertures, all from the table
pass	every land has an aperture rule	all lands ruled
pass	every aperture is above the 0.66 area-ratio floor at 0.12 mm	worst 1.53 at (0.95, 0.6) mm
note	mechanical fit	cadquery is not installed, so the HM-04/HM-05B bayonet fit was not measured on this run
note	frame section	cadquery is not installed, so the HM-01P channel section was not measured on this run
pass	the firmware release images are in the package	four images and a manifest
pass	every image matches the SHA-256 in its manifest	4 images

	Check	Detail
note	firmware release	Built against a real ESP-IDF for the first time on 2026-09-02. Getting here took five corrections to the project, each recorded in the file it was made in: a component that does not exist at this IDF version, a duplicated anti-rollback key that contradicted the Phase 1 intent and made the build refuse a table with a factory partition, a TinyUSB option name that does not exist so the vendor interface was silently absent, and three descriptor callbacks defined here AND in esp_tinyusb. NOT RUN ON HARDWARE: no board exists. This is a build, not a bring-up. Re-issued 2026-09-02 after FW-D17 and FW-D18: the contact-light driver now derives its three colours from two lead-off comparator thresholds rather than from LOFF_STATP and LOFF_STATN, because this board’s montage is single-ended and the N half carried no per-site information, which had made the red state unreachable; and CMD_LIGHTS modes 2, 3 and 4 are implemented rather than silently answered OK.
value	firmware image	405,541 bytes; static IRAM 16,383 used, 1 free

	Check	Detail
open	static IRAM reports full, with one byte free	1 byte(s) free of 16,384 on the linked image. If that is the real limit then the next function marked IRAM_ATTR fails the link with an error naming a section and not a cause, and this design has more interrupt work coming (E-13's tone scheduler, E-12's onset detector). Turning off the SPI-slave and gptimer ISRs – neither of which this firmware uses – did not move the figure by a single byte, so it is not those. Either the pool is genuinely full, or esp_idf_size is reporting against a fixed 16 KB window that is not the limit on an ESP32-S3 with octal SPIRAM and XIP. Reading the linker map against hardware settles it; guessing at sdkconfig from here does not
pass	every printed part is a single closed body	29 meshes, all one body
open	the v1 HM-01 mesh is two disconnected bodies	measured with trimesh: it is watertight, which MANIFEST.json records, and it is not connected, which nothing recorded. It is a carried-over form study rather than a released printable part – the parametric channel runs HM-01P-A/B/C were measured from it – but a frame that is two shells is one more reason the redraw of PARTS-EEG-019 OA-1 has to happen
pass	cross-language interop harness supplied	
pass	layer map and checksums supplied	
pass	Rev B fabrication drawing retained as history	

	Check	Detail
pass	the Rev B geometry has been regraded under the ECO-EEG-032 rule set	
pass	the layout rule sheet is issued	
pass	the 3D collision check has been run	
open	there is no fabrication data for the current board	Rev B is withdrawn from fabrication (ECO-EEG-030) and Rev C is UNROUTED: its placement and routing are bought. The Gerbers, drill, DRC report and drawings this station checks are Rev B's and are kept as history. A bare board cannot be ordered. What exists for Rev C is the layout input set in kicad/RevC_layout_inputs/ and the rule sheet LAY-EEG-034
pass	the Rev C layout input set is assembled	
pass	the Rev C schematic matches the design source	tools/sch_netlist.py reads the emitted .kicad_sch back and diffs it against design.py
pass	every footprint is audited against its part number	docs/footprint_audit_RevC.md
pass	Rev B DRC report retained as history	
pass	the Rev B DRC reported no violations against ITS rule set	VIOLATIONS: 0. That rule set did not contain six of the seven findings of ECO-EEG-030; the same geometry regraded under them is in kicad/EEG-CAR-01_RevB_regraded_ECO-EEG-032.txt
pass	every routable net is one connected copper island	145 of 145 connected, 0 unclosed, 0 without copper
value	nets fully connected	145 of 145 routable (156 in the design)
value	connections routed at relaxed geometry	169

	Check	Detail
note	release state	the ECO-EEG-016 section 3 gate is MET – zero violations, every net one connected copper island. The fabrication data is RELEASED FOR REVIEW under RFQ-EEG-002A and is NOT released for fabrication: no human layout engineer has read this routing, and 169 of its connections close at the minimum conductor or the minimum gap rather than the preferred width
open	the placement of Rev C has not been reviewed	A human layout engineer HAS now read the Rev B routing: he read it between 3 and 5 September 2026, declined the paid review and advised a redesign, and Rev B is withdrawn (ECO-EEG-030). What is open is the next review, not that one. RFQ-EEG-002A is re-scoped to the review of the layout contractor's PLACEMENT of Rev C, taken at the placement confirmation gate and before routing is confirmed, as a written findings list by net, pad pair and coordinate. There is no placement to review yet

Station 02 bare-board electrical test

	Check	Detail
pass	IPC-D-356A netlist supplied	
pass	every net appears in the test netlist	
pass	netlist terminates with 999	
value	test points in the netlist	1172

Station 03 SMT assembly

	Check	Detail
pass	SMT pick-and-place file supplied	

	Check	Detail
pass	every surface-mount part is in the CPL	missing: []
pass	CPL origin is the bottom-left corner, Y positive	x 12.0..145.0 y 7.0..126.7
pass	every CPL row states the layer	
value	SMT placements	153 placed parts, of which R89 is do-not-populate; the CPL has 156 rows because it also carries the 3 fiducials, which take copper and mask apertures but no paste
value	distinct SMT packages	8: C_0603_1608Metric, Fiducial_1mm_Mask3mm, L_0603_1608Metric, R_0603_1608Metric, R_1206_3216Metric, SOT-23, SOT-23-5, TSSOP-14_4.4x5mm_P0.65mm
pass	stencil (paste) layer supplied	
pass	every surface-mount pad that takes paste has a stencil aperture	stencil 366, pads taking paste 366
value	surface-mount pads deliberately without paste	24 (TP1-TP18 probe pads and the three fiducials: copper and mask only, nothing is soldered to them)
note	reflow profile	SAC305: preheat 150-200 C over 90 s, 45-90 s above the 217 C liquidus, peak 235-245 C measured at the U1 body and 245 C maximum anywhere on the board, cooling <= 4 C/s – ASM-EEG-007 section 2.5
pass	assembly work instructions present	

Station 04 through-hole assembly

	Check	Detail
pass	through-hole position file supplied	
pass	every through-hole part is listed	missing: []
value	through-hole parts	33

	Check	Detail
pass	R90 and R91 exist and are the only star points	
pass	no component other than R90 bridges AGND_REF and DGND	bridging parts: [‘R90’]
pass	no component other than R91 bridges HARN_SHIELD and DGND	bridging parts: [‘R91’]

Station 05 module preparation and the MP-01 plate

	Check	Detail
pass	interface control document present	
pass	module plate MP-01 model present	
pass	J1 exists on the carrier	ADS1299 module #1 digital
pass	J2 exists on the carrier	ADS1299 module #1 analogue signals
pass	J23 exists on the carrier	ADS1299 module #1 analogue rails
pass	J3 exists on the carrier	ADS1299 module #2 digital
pass	J4 exists on the carrier	ADS1299 module #2 analogue signals
pass	J29 exists on the carrier	ADS1299 module #2 analogue rails
pass	J6 exists on the carrier	ESP32-S3-DevKitC-1 row A
pass	J7 exists on the carrier	ESP32-S3-DevKitC-1 row B
pass	J8 exists on the carrier	ES8388 codec module
pass	J9 exists on the carrier	Codec microphone feeds
pass	J10 exists on the carrier	ADuM4160 USB isolator, device side
pass	J11 exists on the carrier	ATECC608B breakout
pass	J12 exists on the carrier	Charger + fuel gauge module
pass	J19 exists on the carrier	74HC595 contact-light driver
pass	J20 exists on the carrier	microSD breakout, SDMMC 1-bit

	Check	Detail
pass	J21 exists on the carrier	Boom-mic preamp module (part not settled, E-14)
pass	J25 exists on the carrier	Buck-boost module, TPS63020 class
pass	J28 exists on the carrier	Room microphone module
pass	DevKitC-1 header row spacing is 22.86 mm	22.86 mm
pass	J7 position 11 is left unconnected (PSRAM / strapping)	NC_GPIO37
pass	J7 position 12 is left unconnected (PSRAM / strapping)	NC_GPIO36
pass	J7 position 13 is left unconnected (PSRAM / strapping)	NC_GPIO35
pass	J7 position 15 is left unconnected (PSRAM / strapping)	NC_GPIO45

Station 06 harness and cable assembly

	Check	Detail
pass	harness specification present	
pass	electrode harness carries 8 scalp + 2 reference + bias + screen = 12 ways	E_Fz, E_Cz, E_Pz, E_C3, E_C4, E_T7, E_T8, E_F7, REF_L, REF_R, BIAS_EL, HARN_SHIELD
pass	light harness carries 8 lines + common + return = 10 ways	LED1, LED2, LED3, LED4, LED5, LED6, LED7, LED8, LED_V, LED_GND
pass	no digital signal shares the electrode harness	

Station 07 mechanical and enclosure

	Check	Detail
pass	HM-04_electrode_assembly_body.stl present	
pass	HM-04_electrode_assembly_body.step present (a mesh cannot be dimensioned)	
pass	HM-08_battery_hatch.stl present	

	Check	Detail
pass	HM-08_battery_hatch.step present (a mesh cannot be dimensioned)	
pass	HM-09_service_key.stl present	
pass	HM-09_service_key.step present (a mesh cannot be dimensioned)	
pass	HM-02_brow_pad.stl present	
pass	HM-02_brow_pad.step present (a mesh cannot be dimensioned)	
pass	FIT-01_fit_test_coupon.stl present	
pass	FIT-01_fit_test_coupon.step present (a mesh cannot be dimensioned)	
pass	POD-P1_prototype_enclosure_base.stl present	
pass	POD-P1_prototype_enclosure_base.step present (a mesh cannot be dimensioned)	
pass	POD-P1_prototype_enclosure_lid.stl present	
pass	POD-P1_prototype_enclosure_lid.step present (a mesh cannot be dimensioned)	
pass	MP-01_module_plate.stl present	
pass	MP-01_module_plate.step present (a mesh cannot be dimensioned)	
pass	the CASE-00 Rev C foam cut files are supplied, all seven layers	7 layer file(s) in mech/
pass	no superseded Rev B foam file is still shipped	PKG-EEG-015 2.4: the Rev B two-layer insert cannot pack this kit
pass	2D drawings supplied for the printed parts	
pass	rulings register present	

	Check	Detail
pass	part identifier register present	
pass	carrier fits the POD-P1 internal envelope (158 x 138 mm)	board 150.0 x 130.0 mm
pass	the electronics stack fits the 55.5 mm internal depth	49.1 mm: floor 2.5 + boss 6.0 + carrier 1.6 + standoff 18.0 + plate 3.0 + modules 18.0
value	mounting hole pattern	M3 at (5,5) (145,5) (5,125) (145,125) – 140 x 120 mm centres, shared by the carrier, MP-01 and the POD-P1 bosses

Station 08 firmware build, flash and provisioning

	Check	Detail
pass	firmware/main/main.c present	
pass	firmware/CMakeLists.txt present	
pass	firmware/main/CMakeLists.txt present	
pass	firmware/sdkconfig.defaults present	
pass	firmware/partitions.csv present	
pass	firmware/README.md present	
pass	provisioning script present	
pass	host verification tool present	
pass	generated pin map header present	
pass	no signal is assigned to GPIO35 (octal PSRAM)	
pass	no signal is assigned to GPIO36 (octal PSRAM)	
pass	no signal is assigned to GPIO37 (octal PSRAM)	
pass	no signal is assigned to GPIO45 (VDD_SPI strapping pin)	
pass	the reserved pins are named in the header so nobody re-uses them	

	Check	Detail
value	GPIOs assigned	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 38, 39, 40, 41, 42, 43, 44, 46, 47, 48
note	firmware status	never compiled against a real ESP-IDF installation and never run on hardware; five drivers are stubs (DSN-EEG-003 Rev D section 5)

Station 09 functional test TST-EEG-004

	Check	Detail
pass	test specification present	
pass	test fixture design present	
value	layer count	4 – L1 signal, L2 and L3 reference planes, L4 signal
value	input RC corner	234 Hz
pass	E-10: response at 100 Hz is inside +/-1.0 dB	-0.728 dB with the 68 k fitted, against the +/-1.0 dB branch E-10 states for it
value	Johnson noise of the 68 k series resistor	0.280 uV RMS (0.5-70 Hz)
pass	E-03: total input-referred noise is inside 1.0 uV RMS	0.313 uV RMS
value	envelope filter	f0 = 49.9 Hz, Q = 0.74, -3 dB at 52.2 Hz
value	envelope filter over the C0G tolerance	f0 = 47.5 to 52.5 Hz (-4.8 % to +5.3 %) with C21/C41/C61 and C22/C42/C62 at +/-5 %; Q unchanged, it is set by C2/C1 alone
pass	T12e: the calculated envelope corner is inside TST-EEG-004's 42 to 58 Hz	f0 = 49.9 Hz nominal, measured and recorded per unit
pass	E-11: the envelope corner holds 50 Hz +/-10 % over the capacitor tolerance	f0 spans 47.5 to 52.5 Hz against a 45.0 to 55.0 band

	Check	Detail
value	envelope AC-coupling corner	1.59 Hz
pass	E-11 as restated by ECO-EEG-027: AC corner at or below 2 Hz	1.59 Hz with C20/C40/C60 = 10 uF into 10 kOhm
value	envelope output divider	x0.0909
pass	E-11: a 1.1 V peak envelope lands inside +/-100 mV	100.0 mV
value	comparator threshold	52.1 mV
pass	E-12: the comparator trips within the envelope’s working range	52.1 mV
value	contact-light current per site	1.30 mA
pass	E-27, current only: GPIO48 can source all eight lights	10.4 mA total against a 40 mA rating
pass	E-27: the bicolour phase driver exists in the firmware	lights_phase() alternates LED_V against the shift register
pass	E-27: all three colours are reachable from what ads_init() enables	green, amber and red are computed from g_loff_insens, g_loff_sens, and the converter is configured to feed a swept LOFF_SENSP comparator
open	E-27 has never been seen to light	the driver is written and the current budget is met, but no unit exists, so no light has been driven and TST-EEG-004 T11 – which reads the R/G ratio with a colorimeter – has not been run. The alternation also quantises to the FreeRTOS tick, about 250 Hz rather than exactly 240, which meets the ‘above 100 Hz’ E-27 is written against and is what T11 will actually measure
value	raw sample payload at 1000 Hz	50.0 kB/s (16 channels x 3 bytes + 2 aux, 1000 times a second)
value	framed stream at 1000 Hz	50.8 kB/s (1014 bytes of frame, 1015 after COBS, one every 20 ms)

	Check	Detail
pass	the framed rate matches the 50.7 kB/s ruled in RUL-EEG-021 section B	50.75 kB/s
pass	E-20: one-bit SDMMC has headroom at 1000 Hz	50.8 kB/s needed against about 2000 kB/s available
pass	F-12: full-speed USB carries three times the stream	152 kB/s against about 1200 kB/s usable
value	ring buffer depth at 1000 Hz	126 s of raw samples (6,291,456 bytes at 50.0 kB/s); 124 s if counted over the framed stream
pass	F-06 as relaxed by ECO-EEG-025: at least 90 s of ring	124 s on the pessimistic framed count; the microSD copy covers anything longer
note	the 118 s figure is withdrawn	it divided 6 MB decimal by the framed rate; the ring is 6 MiB and it holds raw samples, so the depth is 126 s (124 s framed) – ECO-EEG-025
value	recording endurance at 1000 Hz	20.0 h at the 150 mA of TST-EEG-004 T3, but only 6.8 h at the 440 mA that ICD-EEG-006 section 2.7 tallies
pass	E-22 on the T3 limit: at least four hours of recording	20.0 h
pass	E-22 also holds at the higher of the two disputed currents	6.8 h at 440 mA, so E-22 is met on either figure
open	the two board-current figures cannot both be right	TST-EEG-004 T3 limits J13 to 150 mA while ICD-EEG-006 section 2.7 tallies about 440 mA. E-22 is met either way (20.0 h against 6.8 h), but the charger, the thermal budget and the T3 limit itself all rest on the number, and it is open item 14 of RFQ-EEG-001 Rev E. It is measured, not calculated, at T3
value	patient auxiliary current, normal condition	0.106 uA (BAV99 leakage dominates)
pass	S-02: normal condition is inside 10 uA DC	0.106 uA
value	single-fault current with a shorted clamp	36.8 uA, limited by the 68 k series resistor

	Check	Detail
pass	S-02: single fault is inside the 50 uA DC limit	36.8 uA with the 68 kOhm series resistor fitted; the input corner is 234 Hz, which E-10 covers
open	SR-01 is closed in the design and not yet signed off	S-02 is now met at 36.8 uA against 50 uA, because ECO-EEG-024 raised R1-R16 to 68 kOhm. Applying the fix the analysis pointed to is not the same as having it approved: the electrical safety reviewer of RISK-EEG-011 section 7 owns SR-01 and that review has not started.

Station 10 software integration with the session runner

	Check	Detail
pass	frame format is fully specified in the RFQ	
value	frame at 1000 Hz	1015 bytes before COBS, one every 20 ms
pass	a 20 ms frame fits comfortably in a full-speed bulk transfer stream	1015 bytes
note	software contract: F-01 composite CDC-ACM + vendor bulk	FW-EEG-001 section 4
note	software contract: F-02 WebUSB and MS OS 2.0 descriptors	FW-EEG-001 section 4
note	software contract: F-04 `iSerialNumber` = the unit serial TIOV-B-nnnn	PKG-EEG-015 section 5 defines the format; RUL-EEG-021 section B rules it
note	software contract: F-08 signed block chain	FW-EEG-001 section 5.6
note	software contract: F-21 timing self-test	TST-EEG-004 step T13
pass	host verification tool covers the signature chain (T16)	

Station 11 final assembly, labelling and kit packing

	Check	Detail
pass	packing, labelling and shipping specification present	

	Check	Detail
pass	participant quick-start card present	
pass	every foam layer the packing list needs is supplied	CASE-00 Rev C, seven 25 mm layers on a 516 x 390 mm sheet

Station 12 despatch

	Check	Detail
pass	regulatory and compliance file present	
pass	lithium shipping is covered (UN3481 / PI967)	
pass	RoHS and REACH declarations are covered	

Station 13 the participant's own handling

	Check	Detail
pass	quick-start card covers: the one counter-intuitive instruction	
pass	quick-start card covers: what a red light means	
pass	quick-start card covers: charging and the wear-while-charging prohibition	
pass	quick-start card covers: the strap is removable and why	
pass	quick-start card covers: the honesty statement	
note	manual handling	fitting is three steps and is expected to take two to three minutes ONCE THE PARTICIPANT HAS DONE IT ONCE (IFU-EEG-014 section 2). A first-timer is allowed up to five minutes before IFU-EEG-014 is treated as wrong (IFU-EEG-014 fit-trial verification; DSN-EEG-002 section 9)

Station 14 return, refurbishment and the next cycle

	Check	Detail
pass	service and refurbishment manual present	
pass	risk analysis covers cross-infection between participants	
pass	change control and document register present	